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SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE		DEPARTMENT OF COMPUTER SCIENCE ENGINEERING	
Program Name: B. Tech		**Assignment Type: Lab**	**Academic Year:**2025-2026
Course Coordinator Name		Dr. Rishabh Mittal	
Instructor(s) Name		Mr. S Naresh Kumar Ms. B. Swathi Dr. Sasanko Shekhar Gantayat Mr. Md Sallauddin Dr. Mathivanan Mr. Y Srikanth Ms. N Shilpa Dr. Rishabh Mittal (Coordinator) Dr. R. Prashant Kumar Mr. Ankushavali MD Mr. B Viswanath Ms. Sujitha Reddy Ms. A. Anitha Ms. M.Madhuri Ms. Katherashala Swetha Ms. Velpula sumalatha Mr. Bingi Raju	
Course Code	23CS002PC304	**Course Title**	AI Assisted Coding
Year/Sem	III/II	**Regulation**	R23
Date and Day of Assignment	**Week3 – Wednesday**	**Time(s)**	23CSBTB01 To 23CSBTB52
Duration	2 Hours	**Applicable to Batches**	All batches
AssignmentNumber:6.3(Present assignment number)/**24**(Total number of assignments)			

Q.No.	**Question**	***Expected Time to complete***
1	**Lab 6: AI-Based Code Completion – Classes, Loops, and Conditionals** **Lab Objectives** • To explore AI-powered auto-completion features for core Python constructs such as classes, loops, and conditional statements. • To analyze how AI tools suggest logic for object-oriented programming and control structures. • To evaluate the correctness, readability, and completeness of AI-generated Python code.	Week3 - Wednesday

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	Lab Outcomes (LOs) After completing this lab, students will be able to: • Use AI tools to generate and complete Python class definitions and methods. • Understand and assess AI-suggested loop constructs for iterative tasks. • Generate and evaluate conditional statements using AI-driven prompts. • Critically analyze AI-assisted code for correctness, clarity, and efficiency.	
	Task Description #1: Classes (Student Class) **Scenario** You are developing a simple student information management module. **Task** • Use an AI tool (GitHub Copilot / Cursor AI / Gemini) to complete a Student class. • The class should include attributes such as name, roll number, and branch. • Add a method display_details() to print student information. • Execute the code and verify the output. • Analyze the code generated by the AI tool for correctness and clarity. **Expected Output #1** • A Python class with a constructor (__init__) and a display_details() method. • Sample object creation and output displayed on the console. • Brief analysis of AI-generated code.	

resume devops.js | Tomato .html | lab 3 devops.html | DOM.html

C: > Users > shash > Downloads > counter-app > AAC A 6.3.py > ...

```python
class Student:
    def __init__(self, name, roll_number, branch):
        self.name = name
        self.roll_number = roll_number
        self.branch = branch

    def display_details(self):
        print(f"Name: {self.name}")
        print(f"Roll Number: {self.roll_number}")
        print(f"Branch: {self.branch}")

if __name__ == "__main__":
    s = Student("Alice Smith", "CS2026", "Computer Science")
    s.display_details()
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
conda : The term 'conda' is not recognized as the name of a cmdlet, func
+ conda activate Shashidhar
+ ~~~~~
    + CategoryInfo          : ObjectNotFound: (conda:String) [], CommandN
    + FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\shash\Downloads\counter-app>  & 'c:\Users\shash\anaconda3\env
\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\de
hash\Downloads\counter-app\AAC A 6.3.py'
Name: Alice Smith
Roll Number: CS2026
Branch: Computer Science
PS C:\Users\shash\Downloads\counter-app>
```

Task Description #2: Loops (Multiples of a Number)

Scenario

You are writing a utility function to display multiples of a given number.

Task

- Prompt the AI tool to generate a function that prints the first 10 multiples of a given number using a loop.
- Analyze the generated loop logic.
- Ask the AI to generate the same functionality using another controlled looping structure (e.g., while instead of for).

Expected Output #2

- Correct loop-based Python implementation.
- Output showing the first 10 multiples of a number.
- Comparison and analysis of different looping approaches.

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```
JS resume devops.js    <> Tomato .html    <> lab 3 devops.h
C: > Users > shash > Downloads > counter-app > AAC A 6.3.py >
def print_multiples_for(n):
    for i in range(1, 11):
        print(n * i)

def print_multiples_while(n):
    i = 1
    while i <= 10:
        print(n * i)
        i += 1

if __name__ == "__main__":
    number = 7
    print("For loop multiples:")
    print_multiples_for(number)
    print("While loop multiples:")
    print_multiples_while(number)
```

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS C:\Users\shash\Downloads\counter-app>  c:; cd 'c:
da3\envs\Shashidhar\python.exe' 'c:\Users\shash\.vsc
libs\debugpy\launcher' '56132' '--' 'c:\Users\shash\
For loop multiples:
28
35
42
49
56
63
70
PS C:\Users\shash\Downloads\counter-app>
```

Task Description #3: Conditional Statements (Age Classification)

Scenario
You are building a basic classification system based on age.

Task
- Ask the AI tool to generate nested if-elif-else conditional statements to classify age groups (e.g., child, teenager, adult, senior).
- Analyze the generated conditions and logic.
- Ask the AI to generate the same classification using alternative conditional structures (e.g., simplified conditions or dictionary-based logic).

Expected Output #3
- A Python function that classifies age into appropriate groups.
- Clear and correct conditional logic.
- Explanation of how the conditions work.

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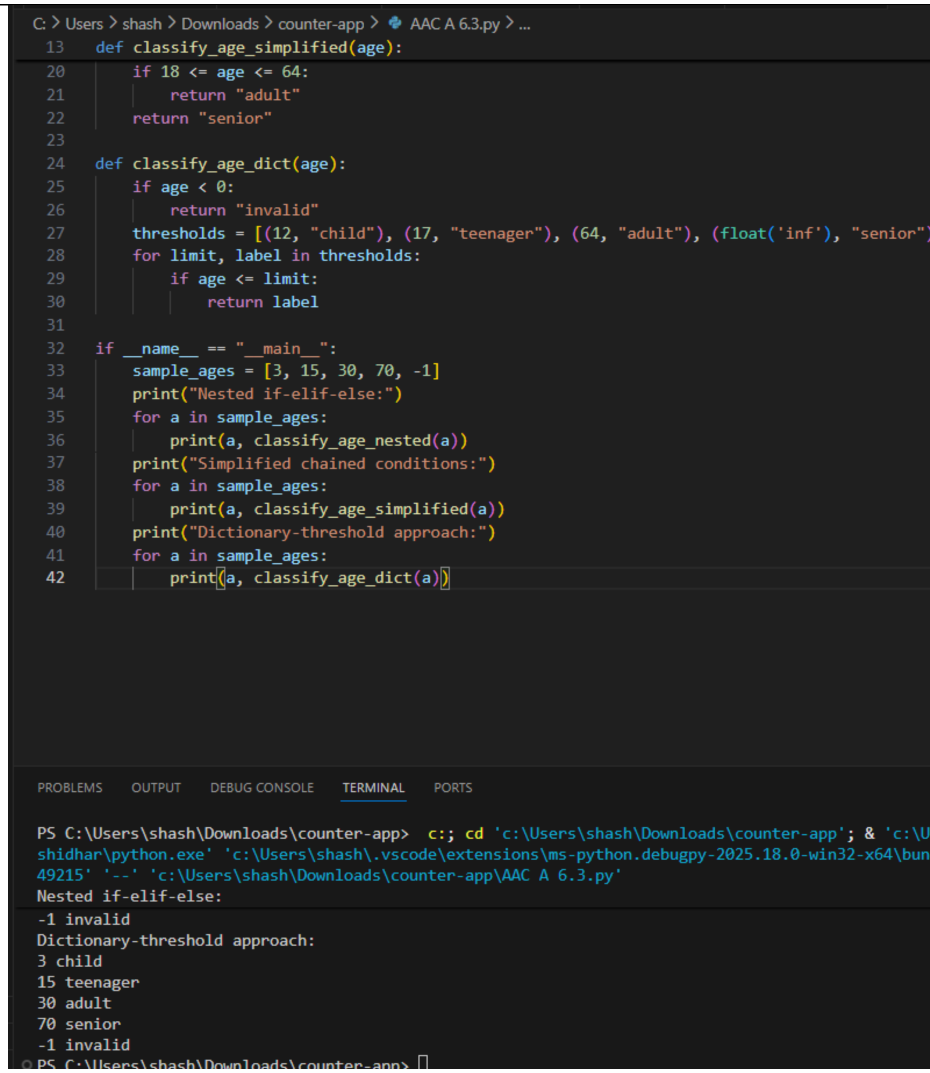
```
C: > Users > shash > Downloads > counter-app > AAC A 6.3.py > ...
def classify_age_nested(age):
    if age < 0:
        return "invalid"
    if age <= 12:
        return "child"
    elif age <= 17:
        return "teenager"
    elif age <= 64:
        return "adult"
    else:
        return "senior"

def classify_age_simplified(age):
    if age < 0:
        return "invalid"
    if 0 <= age <= 12:
        return "child"
    if 13 <= age <= 17:
        return "teenager"
    if 18 <= age <= 64:
        return "adult"
    return "senior"

def classify_age_dict(age):
    if age < 0:
        return "invalid"
    thresholds = [(12, "child"), (17, "teenager"), (64, "adult"), (float('inf'), "senior")]
    for limit, label in thresholds:
        if age <= limit:
            return label

if __name__ == "__main__":
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\shash\Downloads\counter-app>  c:; cd 'c:\Users\shash\Downloads\counter-app'; & 'c:\Users\sh
shidhar\python.exe' 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\li
49215' '--' 'c:\Users\shash\Downloads\counter-app\AAC A 6.3.py'
Nested if-elif-else:
-1 invalid
Dictionary-threshold approach:
3 child
15 teenager
30 adult
70 senior
-1 invalid
PS C:\Users\shash\Downloads\counter-app> 
```



Task Description #4: For and While Loops (Sum of First n Numbers)

Scenario

You need to calculate the sum of the first n natural numbers.

Task

- Use AI assistance to generate a sum_to_n() function using a for loop.
- Analyze the generated code.
- Ask the AI to suggest an alternative implementation using a while loop or a mathematical formula.

Expected Output #4

- Python function to compute the sum of first n numbers.
- Correct output for sample inputs.
- Explanation and comparison of different approaches.

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C: > Users > shash > Downloads > counter-app > AAC A 6.3.py > ...

```python
def sum_to_n_for(n):
    total = 0
    for i in range(1, n + 1):
        total += i
    return total

def sum_to_n_while(n):
    total = 0
    i = 1
    while i <= n:
        total += i
        i += 1
    return total

def sum_to_n_formula(n):
    if n < 0:
        return None
    return n * (n + 1) // 2

if __name__ == "__main__":
    samples = [0, 1, 10, 100]
    for n in samples:
        print(n, sum_to_n_for(n), sum_to_n_while(n), sum_to_n_formula(n))
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\shash\Downloads\counter-app>  c:; cd 'c:\Users\shash\Downloads\coun
da3\envs\Shashidhar\python.exe' 'c:\Users\shash\.vscode\extensions\ms-python.de
libs\debugpy\launcher' '65013' '--' 'c:\Users\shash\Downloads\counter-app\AAC A
0 0 0 0
PS C:\Users\shash\Downloads\counter-app>  c:; cd 'c:\Users\shash\Downloads\coun
da3\envs\Shashidhar\python.exe' 'c:\Users\shash\.vscode\extensions\ms-python.de
libs\debugpy\launcher' '61058' '--' 'c:\Users\shash\Downloads\counter-app\AAC A
0 0 0 0
1 1 1 1
10 55 55 55
100 5050 5050 5050
PS C:\Users\shash\Downloads\counter-app>
```

Task Description #5: Classes (Bank Account Class)

Scenario
You are designing a basic banking application.

Task
- Use AI tools to generate a Bank Account class with methods such as deposit(), withdraw(), and check_balance().
- Analyze the AI-generated class structure and logic.
- Add meaningful comments and explain the working of the code.

Expected Output #5
- Complete Python Bank Account class.
- Demonstration of deposit and withdrawal operations with updated balance.
- Well-commented code with a clear explanation.

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ab 3 devops.html DOM.html CP A 4.7.java AAC A 6.3.py DOM.css

C: > Users > shash > Downloads > counter-app > AAC A 6.3.py > BankAccount > withdraw

```python
class BankAccount:
    def __init__(self, owner, balance=0.0):
        self.owner = owner
        self.balance = float(balance)
    def deposit(self, amount):
        if amount <= 0:
            raise ValueError("Deposit amount must be positive")
        self.balance += amount
        return self.balance
    def withdraw(self, amount):
        if amount <= 0:
            raise ValueError("Withdrawal amount must be positive")
        if amount > self.balance:
            return False
        self.balance -= amount
        return True
    def check_balance(self):
        return self.balance

    def __repr__(self):
        return f"BankAccount(owner={self.owner!r}, balance={self.balance:.2f})"


if __name__ == "__main__":
    owner = input("Enter account owner name: ").strip()
    bal = input("Enter starting balance (leave empty for 0): ").strip()
    try:
        start_balance = float(bal) if bal else 0.0
    except ValueError:
        start_balance = 0.0
    acct = BankAccount(owner or "Unknown", start_balance)
    print("Account created:", acct)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\shash\Downloads\counter-app>  c:; cd 'c:\Users\shash\Downloads\counter-app'
da3\envs\Shashidhar\python.exe' 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-20
libs\debugpy\launcher' '63018' '--' 'c:\Users\shash\Downloads\counter-app\AAC A 6.3.py'
Enter account owner name: Shashidhar Ashadapu
Enter starting balance (leave empty for 0): 1000
Account created: BankAccount(owner='Shashidhar Ashadapu', balance=1000.00)

Options: [d]eposit, [w]ithdraw, [c]heck balance, [q]uit
Choose option: w
Amount to withdraw: 500
Success: True Balance: 500.0
```

Note: Report should be submitted as a word document for all tasks in a single document with prompts, comments & code explanation, and output and if required, screenshots.